

Exp. 7

Qualitative and quantitative determination of carbonate mixtures: method of using two different indicators and the titration using pH meter

The alkalinity of natural waters is usually controlled by OH^- , CO_3^{2-} , and HCO_3^- . In this experiment, the basic principle is to find the composition of a mixture of sodium hydroxide, sodium carbonate, or sodium bicarbonate in unknown sample which may be present singularly or in combination. This is done in two ways: acid-base titration using two different indicators and the pH titration using a pH meter. This is to compare the two methods.

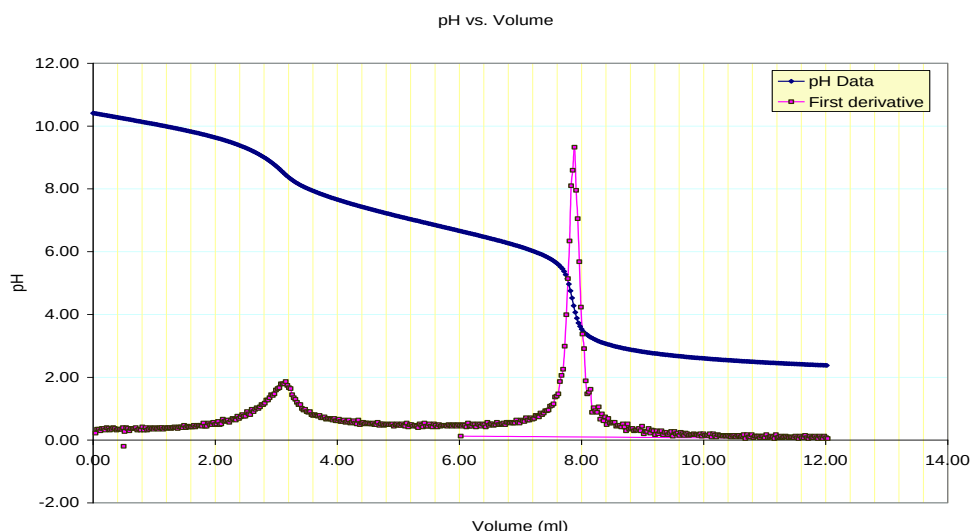
A titration curve is obtained when the pH is measured and plotted against the volume of titrant added. Titration curves illustrate a large change in pH near the equivalence point. The equivalence point is the mid-point of the most vertical section of the titration curve. The equivalence point can be obtained visually or by additional graphing procedure.

The unknown is titrated with standard HCl using a potentiometric (pH) end point measured with a pH meter and combination electrode. The indicator (phenolphthalein, methyl orange) end point results are compared with potentiometric end point results to observe the clarity and variability of various techniques.

The relevant reactions and end point changes are as follows:

<u>Reaction</u>	<u>pH</u>	<u>Indicator used</u>
$\text{OH}^- + \text{H}^+ \rightarrow \text{H}_2\text{O}$	(First end point, pH = 7)	Any indicator
$\text{CO}_3^{2-} + \text{H}^+ \rightleftharpoons \text{HCO}_3^-$	(First end point, pH = 8.3)	Phenolphthalein
$\text{HCO}_3^- + \text{H}^+ \rightleftharpoons \text{H}_2\text{O} + \text{CO}_2$	(Second end point, pH = 4.5)	Methyl orange

pH titration curve illustrates the first and second endpoints:



The purpose of the titration is to locate approximately each of the two equivalence points in the titration of unknown with HCl.

Procedure:

a) Titration using two different indicators

1. Pipet 25 ml aliquot of the sample (V_s) in a conical flask.
2. Introduce 2-3 drops of phenolphthalein solution, and titrate with standardized HCl solution until the solution just begins to change color from pink to colorless. Read the burette mark (V_{pth}).
3. Pipet another 25 ml aliquot of the sample (V_s) in a conical flask.
4. Introduce 2-3 drops of methyl orange solution, and titrate with standardized HCl solution until the solution just begins to change color from yellow to orange. Read the burette mark (V_{mo}).
6. Repeat the titration and calculate the average volumes of used HCl solution.
7. Identify the components of your unknown.
8. Calculate the concentration of components in the sample.



b) Titration using a pH meter

- 1) Calibrate the pH meter using two standard pH buffers (pH 4.00 and pH 7.00).
- 2) Transfer using a volumetric pipet a 25.00 mL aliquot of the sample solution to a 250 mL beaker containing a magnetic stirring bar.
- 3) Place the beaker on a stirrer, immerse the pH electrode, and start the stirrer being careful not to touch the electrode to the stirring bar.
- 4) Titrate with standard HCl, taking pH readings every **2 ml addition** of HCl at the beginning then take readings every **0.5 ml** interval near the equivalence point.
- 5) Plot a curve (using graph paper) of pH versus volume of HCl added.
- 6) Locate the approximate end points (both of them) on the graph.

Qualitative Analysis

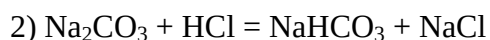
The composition of the solution can be deduced from the relative volumes of acid needed to titrate equal volumes of the sample.

Components in sample	No. of end points	Relationship between V_1 and $V_2 - V_1$ in the titration of an equal volume of sample
NaOH	One (at pH > 7)	V_1
Na ₂ CO ₃	two	$V_1 = V_2 - V_1$
NaHCO ₃	One (at pH < 7)	V_2
NaOH + Na ₂ CO ₃	two	$V_1 > V_2 - V_1$
Na ₂ CO ₃ + NaHCO ₃	two	$V_1 < V_2 - V_1$

Quantitative Calculation

a) If the mixture is sodium hydroxide and sodium carbonate:

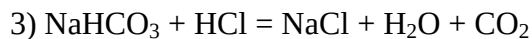
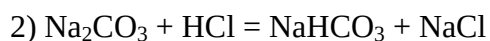
- Titration with hydrochloric acid to a phenolphthalein end point (V_{pth}):



The pink color of the indicator is discharged when reactions 1 and 2 are complete (1st end point)

$$\text{mols of HCl used} = (V_1 \times M_{\text{HCl}}) = \text{mols of OH}^- + \text{mols of CO}_3^{2-} \dots (1)$$

- Titration with hydrochloric acid to a methyl orange end point (2nd end point):



The color of the indicator is changed from yellow to orange when reactions 1, 2, and 3 are complete (2nd end point).

$$\text{mols of HCl used} = \text{mols HCl used for OH}^- + \text{mols HCl used for CO}_3^{2-} + \text{mols HCl used for HCO}_3^-$$

From reaction 2: $\text{mols HCl used for CO}_3^{2-} = \text{mols HCl used for HCO}_3^-$

mols of HCl used = $(V_1 \times M_{\text{HCl}})$ = mols HCl used for OH^- + mols HCl used for CO_3^{2-}

mols of HCl used = $(V_2 \times M_{\text{HCl}})$ = mols of OH^- + 2 mols of CO_3^{2-} (2)

Solve equation 1 & 2 by substitution to find mols of OH^- and mols of CO_3^{2-}

$[\text{OH}^-]$ = mols of OH^- / Volume of sample

$[\text{CO}_3^{2-}]$ = mols of CO_3^{2-} / Volume of sample

b) If the mixture is sodium carbonate and sodium bicarbonate:

- Titration with hydrochloric acid to a phenolphthalein end point (1st end point):

1) $\text{Na}_2\text{CO}_3 + \text{HCl} = \text{NaHCO}_3 + \text{NaCl}$

The pink color of the phenolphthalein indicator is discharged when reaction 1 is complete (1st end point).

mols of HCl used = $(V_1 \times M_{\text{HCl}})$ = mols HCl used for CO_3^{2-}

mols of HCl = mols of CO_3^{2-} (1)

$[\text{CO}_3^{2-}]$ = mols of CO_3^{2-} / Volume of sample

- Titration with hydrochloric acid to a methyl orange end point (2nd end point):

1) $\text{Na}_2\text{CO}_3 + \text{HCl} = \text{NaHCO}_3 + \text{NaCl}$

2) $\text{NaHCO}_3 + \text{HCl} = \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ * NaHCO_3 produced from **reaction 1** by 1:1 ratio

3) $\text{NaHCO}_3 + \text{HCl} = \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ NaHCO_3 existed in the solution

The methyl orange end point indicates the complete conversion of carbonate and bicarbonate to carbonic acid (pH about 4-5). The color of the indicator is changed from yellow to orange when reactions 1, 2, and 3 are complete.

mols of HCl used = mols HCl used for CO_3^{2-} + mols HCl used for * HCO_3^- + mols HCl used for HCO_3^-

From reaction 1: mols HCl used for CO_3^{2-} = mols HCl used for * HCO_3^-

mols of HCl used = $(V_2 \times M_{\text{HCl}})$ = 2 mols of CO_3^{2-} + mols of HCO_3^- (2)

Since mols CO_3^{2-} is known from equation 1, mols HCO_3^- can be calculated.

mols HCO_3^- = $(V_{\text{mo}} \times M_{\text{HCl}}) - 2 \text{ mols } \text{CO}_3^{2-}$

$$[\text{CO}_3^{2-}] = \text{mols of HCO}_3^- / \text{Volume of sample}$$

Example Calculation

You are given a mixture containing NaHCO_3 and Na_2CO_3 . You took 3.5000 g of the sample and dissolved to 250.0 mL in a volumetric flask with deionized water. A 25.0 mL aliquot of the sample required 36.0 mL of standard 0.100 M HCl to reach the methyl orange end point. Another 25.0 mL aliquot was mixed with 50.0 mL of standard 0.100 M NaOH. Then the solution is treated with BaCl_2 to precipitate carbonate as $\text{BaCO}_3(\text{s})$ and immediately titrated with standard 0.100 M HCl to phenolphthalein end point. The volume of HCl required was 40.0 mL. Calculate the wt% of NaHCO_3 and Na_2CO_3 in the mixture.

$$V_{\text{HCl}} (\text{back titrated}) = 40.0 \text{ mL}$$

$$V_{\text{NaOH}} = 50.0 \text{ mL}$$

$$\text{mmols of bicarbonate} = 50.0 \times 0.100 - 40.0 \times 0.100 = 1.00 \text{ mmols}$$

$$\text{mg of NaHCO}_3 = 1.00 \text{ mmol} \times 84 \text{ mg/mmol} = 84.0 \text{ mg (in 25.0 mL aliquot)}$$

$$[\text{HCO}_3^-] = 1.00 / 25.0 = 0.04 \text{ M}$$

$$V_{\text{mo}} = 36.00 \text{ mL}$$

$$36.00 \times 0.1000 = 2 \times \text{mmols of carbonate} + \text{mmols of bicarbonate}$$

$$\text{mmols of carbonate} = 0.5 (3.6 - 1.00) = 1.30$$

$$\text{mg of Na}_2\text{CO}_3 = \text{mmols of carbonate} \times \text{MW of Na}_2\text{CO}_3$$

$$= 1.300 \text{ mmol} \times 106 \text{ mg/mmol} = 137.8 \text{ mg (in 25.0 mL aliquot)}$$

$$[\text{CO}_3^{2-}] = 1.300 \text{ mmols} / 25.00 \text{ mL} = 0.052 \text{ M}$$

$$\text{Wt\% Na}_2\text{CO}_3 = 137.8 \times 100 / 350.0 = 39.4\% \quad \text{Wt\% NaHCO}_3 = 84.0 \times 100 / 350 = 24.0\%$$

QUESTION

Q1. Titrating a 100.0 mL sample to a pH of 8.3 requires 18.70 mL of a 0.0281 M solution of HCl. A second 100.0 mL aliquot requires 48.20 mL of the same titrant to reach pH of 4.5. Calculate the concentrations of CO_3^{2-} and HCO_3^- in ppm.

Q2. a) Calculate the number of moles of sodium carbonate present in a mixture of sodium bicarbonate, sodium carbonate and a neutral component if 6.0 ml of 0.100 M HCl were used to titrate 0.200 g of this mixture with phenolphthalein as the indicator

b) Calculate the mass of sodium carbonate present in the mixture

c) Calculate the number of moles of sodium bicarbonate originally present in the unknown mixture, knowing that 15.0 ml of 0.100 M HCl were used in step 2 of the titration, using methyl orange as the indicator.

Exp. 7

Report sheet

Name:

Section:

Unknown no.:.....

1) The concentration of standardized HCl:

2) The composition of Unknown:

Component 1 is.....

Component 2 is.....

3) Solved equations:

4) The concentration of Component 1 is.....

The concentration of Component 2 is.....

Experiment 8

Precipitation Titration of Chloride

Objectives:

1. To understand the titration theory of chloride with silver nitrate solution
2. To apply Mohr method to the measurement of chloride ion in unknown sample
2. To determine the chloride concentration in natural water

Introduction:

Titration is a common analytical technique in which the concentration of a substance of solution can be determined by adding standard solution that reacts with the unknown. By using the stoichiometry of the reaction and the number of moles of standard solution needed, the concentration of analyte can be determined through titration.

Precipitation titration is a type of titration that based upon silver (I) ion (Ag^+):

* In the **Volhard method** for determination of chloride and other anions:

- A known amount of standard silver nitrate solution is added to the sample solution; the amount of silver nitrate is in excess to react with the halide:



- The excess silver (I) is then back-titrated with standard thiocyanate:

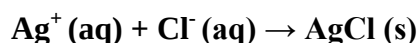


- The solution turns red with first slight excess of thiocyanate ion:



* In **Mohr method**, the most important precipitating agent is silver nitrate in which this method determines the chloride concentration of a solution by titration with silver nitrate, potassium chromate is used as an indicator by reacting with silver ions to produce brick-red silver chromate precipitate at the end point.

As the silver nitrate solution is slowly added, the chloride ion will react with silver ion to form silver chloride precipitate.



$$K_{\text{sp}} = 1.76 \times 10^{-10}$$

This reaction continues until all the chloride ion in the system has been used up to form precipitate and this approaches the end point of titration. After all the chloride ion has been precipitated, any more silver(I) ion will react with the chromate ion, CrO_4^{2-} to form a red-brown precipitate of silver chromate as shown in the following equation.



Endpoints

In any titration, it is necessary to have some method of detecting when just enough of the titrant has been added - a procedure known as detecting the endpoint of the titration. The endpoint of this titration can be detected if the rapid change in either the concentration of silver ion or the concentration of chloride ion which occurs at the endpoint can be made apparent to an observer. In this case we can use Ag_2CrO_4 , because a solution of CrO_4^{2-} is yellow while Ag_2CrO_4 is blood-red.

Suppose $[\text{CrO}_4^{2-}]$ in the solution is 0.001 molar. Then $K_{\text{sp}} = 1.12 \times 10^{-12} = [\text{Ag}^+]^2[\text{CrO}_4^{2-}]$. Since $[\text{CrO}_4^{2-}] = 10^{-3}$ molar, $[\text{Ag}^+]^2 = 1.12 \times 10^{-10}$, and $[\text{Ag}^+] = 3.35 \times 10^{-5}$. At any concentration of silver ion greater than 3.35×10^{-5} in such a solution, a precipitate of Ag_2CrO_4 will form; if the concentration is below this, it will not.

When we have a solution of 0.01 molar Cl^- and add Ag^+ to it, the solubility product is $K_{\text{sp}} = 1.76 \times 10^{-10} = [\text{Ag}^+][\text{Cl}^-]$, so at the start of the titration $[\text{Ag}^+] = 1.76 \times 10^{-10}/1 \times 10^{-2} = 1.76 \times 10^{-8}$ and no precipitate of Ag_2CrO_4 can form.

At the equivalence point, $[\text{Ag}^+] = [\text{Cl}^-]$ and $[\text{Ag}^+]^2 = K_{\text{sp}} = 1.76 \times 10^{-10}$, $[\text{Ag}^+] = 1.33 \times 10^{-5}$ and no precipitate will form. But when a drop or two more of Ag^+ solution is added after the equivalence point has been reached, there is no more Cl^- to react with it. The concentration of silver ion may go up to say 10^{-3} molar. The solubility product of silver chromate will then be exceeded and a red precipitate of Ag_2CrO_4 will designate the end of the titration. This is known as the **Mohr method** of chloride determination.

Mohr method can be used to determine the chloride ion concentration of water samples from many sources such as seawater, stream water, river water and estuary water. The Mohr method works well in a slightly acidic condition, hence the pH of the water sample should be adjusted to the range of 6~8 by adding dilute acid or dilute base as needed.

Example. In a precipitation titration of 46.00 mL of a chloride solution of unknown concentration, 31.00 mL of 0.6973 molar AgNO_3 were required to reach the equivalence point. The molar concentration of the unknown solution is calculated as follows:

$$\begin{array}{rcl} \text{molar} & = & 21.62 \text{ mmol Ag}^+ = 21.62 \text{ mmol Cl}^- \\ \text{mmol Cl}^- / 46.00 \text{ mL Cl}^- & = & 0.4700 \text{ molar Cl}^- \end{array}$$

$31.00 \text{ mL} \times 0.6973$
 21.62

A) Standardization of AgNO_3 solution:

This solution was standardized against NaCl.

- 1) Weigh 0.2500 g of KCl into Erlenmeyer flask and dissolve in about 50 mL of distilled water.
- 2) Add about 5 drops of K_2CrO_4 indicator.
- 3) Titrate the yellow solution with AgNO_3 to the first permanent appearance of red $\text{Ag}_2\text{Cr}_2\text{O}_4$.

B) Determination of Cl^- in solid sample:

- 1) Weigh about 0.2000 g of the unknown sample into 250-mL Erlenmeyer flask and dissolve in about 50 mL of distilled water.
- 2) Add about 5 drops of K_2CrO_4 indicator.
- 3) Titrate the yellow solution with AgNO_3 to the first permanent appearance of red $\text{Ag}_2\text{Cr}_2\text{O}_4$.
- 4- Calculate % Cl^- in the unknown.

C) Determination of chloride content in drinking water

- 1- Pipet 50 ml tap water in a conical flask
- 2- Add 2-3 drops of $\text{K}_2\text{Cr}_2\text{O}_7$
- 3- Titrate the yellow solution with AgNO_3 to the first permanent appearance of red $\text{Ag}_2\text{Cr}_2\text{O}_4$.
- 4- Calculate chloride content (ppm) in water.

Experiment 8- Precipitation titration

Report sheet

Name: _____ Number: _____

1. Standardization of a AgNO_3 Solution with KCl

Mass of KCl (g)=	Trial 1	Trial 2	Trial 3
Volume of titrated sample			
Volume of AgNO_3 used			
Moles of Cl^- in titrated sample			
Moles of Ag^+			
Molarity of AgNO_3			
Average Molarity of NaOH			

2. Determination of the % Cl^-

Unknown no.:	Trial 1	Trial 2	Trial 3
Mass of unknown sample (g)			
Molarity of AgNO_3 used			
Volume of AgNO_3 used			
Moles of Ag^+			
Moles of Cl^-			
Mass of Cl^- (g)			
Mass percent (%)			
Average Mass percent			

3- Determination of chloride in drinking water

	Trial 1	Trial 2	Trial 3
Volume of tap water (L)			
Molarity of AgNO_3 used			
Volume of AgNO_3 used			
Moles of AgNO_3			
Moles of Cl^-			
Mass of Cl^- (mg)			
Chloride content of water (ppm Cl^-)			
Average ppm Cl^-			

Experiment 8

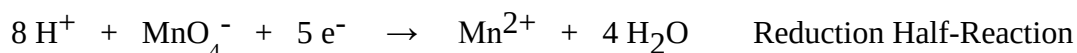
Oxidation-Reduction Titrations

Reactions in which electrons are transferred from one species to another are called oxidation-reduction (redox) reactions. When one species loses electrons by an oxidation process another species simultaneously gains electrons by a reduction process in a chemical reaction. The balanced chemical reaction can be written as the combination of two half-reactions representing the oxidation reaction and the reduction reaction, respectively.

Oxidation is defined as the loss of electrons (on right of arrow) or increase in oxidation state (each C changes from +3 to +4) as shown below by the half-reaction involving oxalate ion.

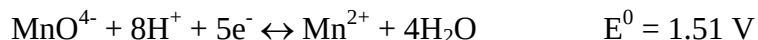


Reduction is shown by the permanganate ion as it gains electrons and Mn decreases in oxidation state from +7 to +2.



Standardization of potassium permanganate with oxalic acid

The permanganate ion is strong oxidizing reagent. The half-reaction is:

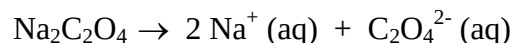


Potassium permanganate is not primary standard substance because it contains reduced products like manganese oxide MnO_2 where the concentration of KMnO_4 changed after preparation because it dissociated via reducing agents such as ammonia and organic substances in water; therefore, potassium permanganate must be standardized before use it and keep it at least 7-10 days after preparation at dark place and dark bottle

Oxalate compounds have importance in many chemical and biological reactions. They are used in the photographic sciences, and are the main component of kidney stones. Excess vitamin C is converted into oxalate and excreted in the gut and urine, therefore, kidney stone patients must limit their intake of vitamin C to control the formation of calcium oxalate stones, as well as foods high in oxalate, such as spinach, beets, and beans.

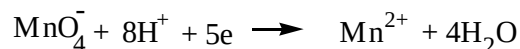
Sodium oxalate is commonly used as a primary standard for determining the concentration of many strong oxidizers used in oxidation/reduction analyses.

Sodium oxalate, $\text{Na}_2\text{C}_2\text{O}_4$, is a strong electrolyte that dissociates completely in water according to the following equation:

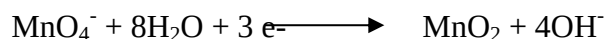


Oxidation process of potassium permanganate conducted in acidic, neutral and basic and it's as follows:

At acidic medium,



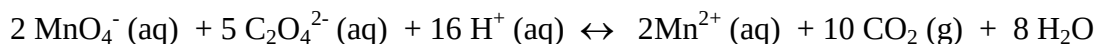
At neutral or basic medium like, Sodium or potassium carbonate



Aqueous solutions of permanganate are not entirely stable because the permanganate ion tends to oxidize water forming manganese dioxide.



In this experiment, the oxalate ion, $\text{C}_2\text{O}_4^{2-}$, will react quantitatively with permanganate ion, MnO_4^- , in the presence of strong acid according to the Oxidation-Reduction reaction:



The numerical value of K_{eq} for the equilibrium is very large. When reaction conditions are optimum (in terms of pH and temperature) the reaction can be considered to be quantitative.

No indicators are necessary in permanganate titrations because the end point is easily observed. The permanganate ion is intensely purple, whereas the manganese (II) ion is nearly colorless. The first slight excess of permanganate imparts a pink color to the solution, signaling that all of the oxalate has been consumed.

Procedure

A) Standardization of potassium permanganate

1. Weigh 0.2 – 0.3 g of sodium oxalate (134 g/mol) and transfer into 100 mL volumetric flask.
2. Dilute to the mark with 1.0 M sulfuric acid.
3. Pipet 25.0 ml in conical flask.
4. shake the solution well and warm it (75-80 °C)
5. Titrate the hot solution against potassium Permanganate solution.

6. Continue with titration drop by drop till the color is changed from colorless to pink
7. Calculate the molarity of MnO_4^- .

B) Determination of oxalate in a mixture

1. Choose an unknown and record its number.
2. Weigh 0.2 – 0.3 g of unknown, transfer into 100 mL volumetric flask, and dilute to the mark with 1.0 M sulfuric acid.
3. Pipet 25.0 ml aliquot in conical flask.
4. shake the solution well and warm it (75-80 °C)
5. Titrate the hot solution against potassium Permanganate solution.
6. Continue with titration drop by drop till the color is changed from colorless to pink
7. Calculate % sodium oxalate in the unknown.

Experiment (10) Redox titration

A) Standardization of potassium permanganate

Wt $\text{Na}_2\text{C}_2\text{O}_4$		
Total mol of $\text{Na}_2\text{C}_2\text{O}_4$		
Volume of titrated sample		
mol $\text{Na}_2\text{C}_2\text{O}_4$ in titrated sample		
Mol KMnO_4		
Volume KMnO_4		
Molarity KMnO_4		
Average molarity		

B) Determination of oxalate in a mixture

Wt unknown		
Volume of titrated sample		
Volume KMnO_4		
Mol KMnO_4		
mol $\text{Na}_2\text{C}_2\text{O}_4$ in titrated sample		
Total mol of $\text{Na}_2\text{C}_2\text{O}_4$		
wt of $\text{Na}_2\text{C}_2\text{O}_4$		
% $\text{Na}_2\text{C}_2\text{O}_4$		
Average % $\text{Na}_2\text{C}_2\text{O}_4$		